Rutgers Health Hack 2025

AWS Resources & Amazon Q Developer Onboarding Guide

1. Setting up Amazon Q Developer

Amazon Q Developer is your AI-powered coding assistant. Setting it up first will also configure AWS Identity Center for your team.

1 Navigate to Amazon Q Developer

Within the AWS Console:

- In the search bar at the top, type `Amazon Q`
- Click on **"Amazon Q Developer"** from the results

2 Create Amazon Q Developer Profile

You'll be prompted to create an Amazon Q Developer profile:

- Enter a **Profile name** (e.g., "QDevProfile-us-east-1")
- Optionally add a profile description to help users identify which profile to select
- The profile will be shared with all associated member accounts
- Click **"Create"**

3 Set up Identity Center

After creating the profile, you'll be prompted to set up Identity Center:

- Select **"Get started with IAM Identity Center"**
- This will automatically configure Identity Center in your workshop account
- Follow the prompts to complete the Identity Center setup

Important: Setting up Amazon Q Developer with Identity Center is a prerequisite for adding team members and using SageMaker Unified Studio later. This creates the foundation for multi-user access.

4 Complete Initial Configuration

- Review the subscription options and choose the appropriate tier
- The free tier is sufficient for getting started
- Confirm your settings

2. Adding Team Members to Identity Center

Now that Identity Center is set up, you can add your teammates so they can access Q Developer & SageMaker Unified Studio (see below).

1 Navigate to Identity Center

Within the AWS Console:

- In the search bar, type `IAM Identity Center`
- Click on **"IAM Identity Center"**

2 Add Individual Users

- In the left navigation panel, choose **"Users"**
- Click **"Add user"**
- Fill in the required information:
  - **Username:** Must be between 1-100 characters
  - **Email address:** Each user needs a unique email
  - **First name** and **Last name**

3 Configure Password Settings

Choose how the user will set up their password:

- **Send an email:** User receives an invitation email from AWS with setup instructions (recommended)
- **Generate a one-time password:** You can share credentials manually

4 Repeat for Each Team Member

Add each teammate individually by repeating the process above.

3. Setting up SageMaker Unified Studio

SageMaker Unified Studio provides cloud-based development environments with Code Editor (based on VS Code) and collaborative spaces for your team.

1 Navigate to SageMaker

Within the AWS Console:

- In the search bar, type `SageMaker`
- Click on **"Amazon SageMaker"**

2 Start Creating a Unified Studio Domain

- Click **"Create a Unified Studio domain"**
- Choose **"Quick setup"** for faster configuration
- This option automatically configures:
  - Data analytics and machine learning capabilities
  - Generative AI app development with Amazon Bedrock
  - Amazon Q integration (free tier)
  - Authentication via AWS IAM Identity Center (already set up!)

3 Create VPC

You must create a VPC before continuing. Here's how:

- In the VPC dropdown, you'll see "No VPC available"
- Click the **"Create a new VPC"** link
- A new browser tab will open with VPC Quick Create
- In the new tab, click **"Create VPC"** to deploy the VPC
- Wait for the VPC creation to complete (takes about 1-2 minutes)
- Once complete, **return to the SageMaker tab**
- Click the **refresh button** next to the VPC dropdown
- Your new VPC will now appear in the dropdown - select it

4 Review Quick Setup Settings

- Expand **"Quick setup settings"** to review configuration
- You can modify domain name, roles, and encryption settings if needed
- Leave defaults for the hackathon, then click **"Continue"**

5 Create Domain Admin User

- On the "Create IAM Identity Center user" page, select your SSO user
- This user will be the administrator in SageMaker Unified Studio
- Click **"Create domain"**

Note: Domain creation takes approximately 5-15 minutes. The status will show "Pending" during creation, then change to "InService" when ready.

Adding Team Members

6 Add Users to the Domain

Once your domain is created:

- Click on your domain name to open the domain dashboard
- Scroll down to the **"User management"** section
- Click **"Add users and groups"**
- Select your team members from the Identity Center users list
- Assign them appropriate access (select **"User"** or **"Power user"** profile)
- Click **"Add"** to grant them access

Accessing Unified Studio

7 Open Unified Studio

- From the domain dashboard, click the **"Open Unified Studio"** button
- A new tab will open
- Click **"Sign in with SSO"**
- Authenticate with your Identity Center credentials
- You'll be taken to the Unified Studio interface

Setting Up Projects and Spaces

8 Create a Project

Projects organize your team's work. You need to create a project before creating spaces:

- In the Unified Studio interface, look for **"Projects"** in the left navigation
- Click **"Create project"**
- Enter a project name (e.g., "HealthHack-Team-Project")
- Add a description for your hackathon project
- Click **"Create project"**

9 Enable Spaces for the Domain

Make sure spaces are enabled for your domain:

- From the project view, look for space-related options
- If prompted to enable spaces, click **"Enable spaces"**
- This allows team members to create individual development environments

10 Create Spaces (One Per Team Member)

Spaces are isolated development environments. **Best practice: Create one space per team member.**

- From your project, click **"Spaces"** in the left navigation
- Click **"Create space"**
- Configure the space:
  - **Name:** Use team member's name (e.g., "johns-workspace")
  - **Application:** This field appears right after Name. **Change it from "JupyterLab" to "Code Editor"** for a VS Code experience (unless you prefer JupyterLab!)
  - **Instance type:** `ml.t3.medium` or `ml.t3.large`
- Click **"Create space"**
- **Repeat for each team member** - this gives everyone their own environment

Tip: Each team member should have their own space to avoid conflicts. Spaces can share code through Git repositories or shared project storage.

11 Launch Your Space

- From the Spaces list, find your space
- Click **"Run space"** to start it (takes 2-3 minutes)
- Once the status shows "Running", click **"Open"**
- The Code Editor will launch in a new tab - it's a full VS Code environment!
- Amazon Q is already integrated and ready to use

4. Installing VS Code Locally

For local development, you'll need to install Visual Studio Code on your machine.

1 Download VS Code

- Visit `code.visualstudio.com`
- Click the download button for your operating system:
  - **Windows:** Download the Windows installer (.exe)
  - **macOS:** Download the macOS installer (.dmg)
  - **Linux:** Download the .deb or .rpm package, or use Snap

2 Install VS Code

Windows:

- Run the downloaded installer
- Accept the license agreement
- Choose installation location (default is fine)
- Select additional tasks (recommend "Add to PATH" and "Create desktop icon")
- Click "Install"

macOS:

- Open the downloaded .dmg file
- Drag Visual Studio Code to the Applications folder
- Launch VS Code from Applications or Spotlight

Linux:

- For Debian/Ubuntu:

```
sudo dpkg -i code_*.deb
sudo apt-get install -f
```

- For Red Hat/Fedora:

```
sudo rpm -i code-*.rpm
```

- Or use Snap:

```
sudo snap install code --classic
```

3 Verify Installation

- Launch VS Code
- You should see the Welcome screen
- VS Code is now ready for extension installation

5. Setting up Amazon Q in Local VS Code

Important: You're using **Amazon Q Developer Pro** (not the free tier). You must authenticate using the **"Company account"** option with your IAM Identity Center Start URL.

1 Install the Amazon Q Extension

- Open VS Code
- Click the Extensions icon in the left sidebar (or press `Ctrl+Shift+X` on Windows/Linux or `Cmd+Shift+X` on Mac)
- In the search box, type `Amazon Q`
- Look for the official extension by Amazon Web Services
- Click **"Install"**

2 Get Your Start URL from Identity Center

You need your Identity Center Start URL (from Section 1 setup):

- Go to the AWS Console and search for `IAM Identity Center`
- On the dashboard, look for the **"AWS access portal URL"**
- This is your **Start URL** (e.g., https://d-xxxxxxxxxx.awsapps.com/start)
- Copy this URL - you'll need it in the next step
- Note the AWS Region shown (e.g., us-east-1)

3 Authenticate with IAM Identity Center

- In VS Code, click the Amazon Q icon in the left sidebar
- **IMPORTANT:** Choose **"Company account"** (NOT "Personal account" or "Builder ID" - those are for the free tier!)
- Paste your **Start URL** from the previous step
- Select your **AWS Region** (where you set up Identity Center)
- Click **"Continue"**
- Your web browser will open for authentication
- Sign in with your Identity Center username and password
- Approve the authorization request
- Return to VS Code - you should now see "Connected" status

4 Start Using Amazon Q

Once authenticated, you can:

- Click the Amazon Q icon to open the chat panel
- Ask questions like "How do I parse JSON in Python?"
- Get inline code suggestions as you type
- Use `Ctrl+I` (or `Cmd+I`) for inline assistance

6. Enabling Amazon Bedrock Models

Amazon Bedrock provides access to foundation models from leading AI companies. You need to request access before using them.

1 Navigate to Amazon Bedrock

Within the AWS Console:

- In the search bar, type `Bedrock`
- Click on **"Amazon Bedrock"**

2 Access Model Access Page

- In the left navigation panel under **"Bedrock configurations"**, click **"Model access"**
- Click the **"Modify model access"** button

3 Select Models to Enable

Review the available models and select the ones you want to use:

- Check the boxes next to the models you want
- You can select individual models or check entire provider groups
- Review each model's End User License Agreement (EULA) before requesting

Cross-Region Inference (CRIS): You can use cross-region inference by adding `us.` to the beginning of a model ID. This enables cross-region inference support, which automatically routes requests across multiple AWS Regions in the US for higher

throughput and better availability. For example, to use Claude Sonnet 4 with cross-region inference, use the model ID that starts with `us.`

4 Submit Access Request

- Click **"Next"**
- Review the terms and pricing information
- Click **"Submit"**

5 Wait for Access Approval

- Access is usually granted instantly or within a few minutes
- The **"Access status"** will change from "In progress" to **"Access granted"**
- Refresh the page to see updated status

7. Using the Bedrock Chat Playground

1 Navigate to Playgrounds

- In the Bedrock console, look for **"Playgrounds"** in the left navigation
- Click **"Chat"** for conversational AI interactions

2 Select a Model

- At the top of the playground, click **"Select model"**
- Choose from your enabled models (e.g., Claude Sonnet 4)
- You can also select cross-region inference profiles for better performance

3 Configure Model Parameters (Optional)

In the right panel, you can adjust:

- **Temperature:** 0.0 for deterministic, 1.0 for creative responses
- **Max tokens:** Maximum length of the response
- **Top P:** Controls diversity via nucleus sampling

4 Start Your Conversation

Type your prompt in the chat box and press Enter. Example prompts for your health hack:

```
"Help me design a database schema for tracking patient vitals"

"Generate Python code to analyze blood pressure trends"

"What are best practices for handling medical data securely?"

"Create a React component for displaying health metrics"
```

8. Launching an EC2 Instance (Optional)

If you need a server for your application backend or database, you can launch an EC2 instance.

1 Navigate to EC2

Within the AWS Console:

- In the search bar, type `EC2`
- Click on **"EC2"**

2 Launch Instance

- Click the orange **"Launch instance"** button
- You'll be taken to the instance configuration page

3 Configure Instance Settings

Name:

- Enter a descriptive name (e.g., "rutgers-health-hack-server")

Operating System:

- Select an OS image (e.g., **"Amazon Linux 2023"**, **"Ubuntu Server 24.04 LTS"**, or **"Windows Server 2025"**)

Instance Type:

- Choose an instance type based on your needs (e.g., `t3.micro`, `t3.small`, `t3.medium`)

4 Create Key Pair

- Click **"Create new key pair"**
- Enter a name (e.g., "rutgers-health-hack-key")
- Select `RSA` key type
- Choose format:
  - `.pem` for Mac, Linux, or Windows with WSL
  - `.ppk` for Windows with PuTTY
- Click **"Create key pair"**
- The file downloads automatically - **save it securely!**

Important: You cannot download the key pair again. Store it safely - you'll need it to connect to your instance.

5 Configure Network Settings

- Keep the default VPC and subnet
- Enable **"Allow SSH traffic from My IP"** (for secure access)
- For web applications, also enable:
  - **"Allow HTTP traffic from the internet"**
  - **"Allow HTTPS traffic from the internet"**

6 Configure Storage

- Default is 8 GB (free tier eligible)
- Increase to 16-30 GB if needed
- Keep `gp3` volume type for good performance

7 Launch Instance

- Review your configuration in the summary panel
- Click **"Launch instance"**
- Wait 1-2 minutes for the instance to start

8 Connect to Your Instance

- Go to **"Instances"** in the left sidebar
- Select your instance
- Click **"Connect"**
- Choose connection method:
  - **EC2 Instance Connect:** Browser-based (easiest)
  - **SSH client:** Using your key pair (most flexible)

For SSH from terminal:

```
chmod 400 rutgers-health-hack-key.pem
ssh -i "rutgers-health-hack-key.pem" ec2-user@[your-instance-public-ip]
```

Replace `ec2-user` with `ubuntu` if you chose Ubuntu Server.

Additional Resources & Support

Documentation

- **Amazon Q Developer:** docs.aws.amazon.com/amazonq
- **Amazon Bedrock:** docs.aws.amazon.com/bedrock
- **SageMaker Unified Studio:** docs.aws.amazon.com/sagemaker/latest/dg/unified-studio.html
- **Amazon EC2:** docs.aws.amazon.com/ec2
- **IAM Identity Center:** docs.aws.amazon.com/singlesignon

Quick Start Checklist

- □ Set up Amazon Q Developer with Identity Center
- □ Add team members to Identity Center
- □ Create SageMaker Unified Studio domain with spaces
- □ Install VS Code locally (optional)
- □ Install Amazon Q extension in VS Code (optional)
- □ Enable Bedrock models (including cross-region ones)
- □ Test the Bedrock Chat Playground
- □ Launch EC2 instance if needed
- □ Start building your health hack project!

Important Information

Workshop Resources & Monitoring:

- Usage of AWS resources during this event is monitored by workshop facilitators
- Workshop resources, including access to Amazon Q Developer, are provided through the completion of the event
- **At the close of the event, all AWS workshop resources will be terminated**
- Please ensure you save any important work or code to your local machine or personal repository before the event ends